



# Big Bang Matters

In the early days, Mega Science was defined as 'big money, big machines' and was used to refer mainly to unique experimental apparatus like particle accelerators, ground or space telescopes like the Hubble Space Telescope, and Space Exploration (ESA and ISS). However, this definition has evolved since and now applies to complex research where not only very large sums of money, necessitating partnerships between different countries, are a requirement but also large teams of competent researchers, thus needing cross-border co-operations between countries and participating institutes often over long periods of time. Consequently an efficient technical coordination and streamlined resource management becomes mandatory over the project duration. The evolution of cross-disciplinary competence is a natural outcome throughout the life of such a Mega Project.

A pioneering example is the European Laboratory for Particle Physics (CERN), in Geneva, Switzerland. India's role in the Large Hadron Collider experimental program at CERN showed the entire world that we can participate in giant endeavours spanning the globe and have the technological wherewithal in both, hardware and software technologies. Nonetheless, no country can keep its progress at par with international standards without the assistance of other countries.

As Bertrand Russel once said: "almost everything that distinguishes the modern world from earlier centuries is attributable to the progress in science". This desire to extend our knowledge and go beyond the present limits is part of our heritage and cultural life. Benefits follow naturally.

A mega project provides opportunities and platforms to work together, facilitating first-hand experience of work ethic and scientific and technological developments in other countries. By allowing access to sophisticated research facilities, it permits an interaction between colleagues at all levels thus filling the lacuna of know-how and knowledge resulting in an accelerated pace of development.

Coming to the point of education and outreach, nothing can beat the 'Black Hole'

phenomena that suddenly glamorized particle physics all over the world, particularly in India. This is one example where a mega science project caught the imagination (remember the imaginary end of the world on September 10, 2008?) and popularised science in the country. A little known CERN Laboratory in Geneva became the buzzword. Several institutes, students and teaching staff from remote corners of the country became interested in forging partnerships, interning and becoming a part of the phenomena.

We need to continue and take these partnerships to another dimension; namely, by bringing the large science into each classroom and into the common man's life.

What are commercial benefits from mega science projects?

As an incubator for invention and for the development of new products, services,

ideas and organizations, cross-disciplinary frontier research results in ambitious initiatives. Did anyone predict the colossal impact of the World Wide Web, developed at CERN for sharing scientific data among collaborators? To propose and defend a bold idea is only possible within a framework of like-minded competent people which is provided by a mega project.

The list of spin-offs is endless. The internet provides seamless access to information that is stored in many different geographical locations. The Grid is an infrastructure that provides seamless access to computing power and data storage capacity distributed over the globe. The weather Grid system comprises thousands of interlinked computers helping to process complex weather data to improve forecasting. Money and human lives can, and are, being saved using accurate weather forecasting.

In a nutshell, significant returns on financial investments are made in mega projects. Financial multipliers in the order of 2.7 for ESA and 3.7 for CERN clearly indicate that money invested in mega science generates two- to four-fold returns for the investors and industry at large.

International collaborations at CERN for example, have been very successful in technology transfer, where research developments have led to applications in other fields. A pioneering study carried out at CERN into the effect of the experience that technological and industrial partners gain through working in the arena of mega science revealed a variety of outcomes which included technological learning, the development of new products and markets and impact on the firm's organization. Together, these findings implied ways in which CERN and by implication other mega science centres and projects - could further boost technology transfer into spill-over benefits for industrial knowledge and enhance their contribution to industrial R&D and innovation.

We look forward to unlocking answers to basic questions on origins of the universe and the technological fallout in the years to come. 



**Archana Sharma**

Archana Sharma is an Indian experimental physicist and a staff member of CERN European Organization for Nuclear Research. She has been working there since 1987 and is presently working on the Compact Muon Solenoid (CMS) experiment at the Large Hadron Collider (LHC) project in Geneva, Switzerland.